Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Srushti Chaudhari
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**AIML Lab**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	6
Course Outcome	CO1: Apply Artificial Intelligence problem formulation techniques, intelligent agents, search algorithms, and logical inference to solve real-world problems. CO2: Implement and evaluate machine learning algorithms including classification, clustering, and dimensionality reduction using appropriate preprocessing and performance metrics. CO3: Analyze and compare ensemble learning models.
Aim	Decision Tree Classifier To build a **Decision Tree classifier** and evaluate its performance using **Confusion Matrix, Precision, Recall, and F1-score**, and analyze the concepts of **overfitting and underfitting**.
Problem Definition	Decision Trees are supervised learning algorithms used for classification and regression tasks. In this practical, a **Decision Tree classifier** is trained using a dataset such as the **Iris dataset**. The model learns decision rules based on feature values to classify data into different categories. After training the model, its performance is evaluated using a **confusion matrix** and evaluation metrics such as **precision, recall, and F1-score**. Additionally, the behavior of the model is analyzed to understand **overfitting**, where the model learns the training data too well, and **underfitting**, where the model fails to capture important patterns in the data.
Theory (100 words)	A Decision Tree is a supervised machine learning algorithm that represents decisions in a tree-like structure. Each internal node represents a feature test, each branch represents the outcome of the test, and each leaf node represents a class label. Decision trees are easy to understand and interpret because they mimic human decision-making processes. The performance of a Decision Tree model is evaluated using metrics such as precision, recall, and F1-score, which are derived from the confusion matrix. However, decision trees can suffer from overfitting if the tree becomes too complex. Techniques such as pruning and controlling tree depth help reduce overfitting.

Procedure and Execution

(100 Words)

Algorithm:

1. Start the program.
2. Import required machine learning libraries.
3. Load the dataset (for example, the Iris dataset).
4. Split the dataset into **training and testing sets**.
5. Train the **Decision Tree classifier** using the training data.
6. Use the trained model to predict class labels for the test data.
7. Generate the **confusion matrix**.
8. Calculate evaluation metrics such as **precision, recall, and F1-score**.
9. Analyze model performance for **overfitting or underfitting**.
10. Display the results and stop the program.

Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix, precision_score,
recall_score, f1_score

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2)

# Train Decision Tree
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Confusion Matrix
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Evaluation Metrics
print("Precision:", precision_score(y_test, y_pred,
average='macro'))
print("Recall:", recall_score(y_test, y_pred, average='macro'))
print("F1 Score:", f1_score(y_test, y_pred, average='macro'))
```

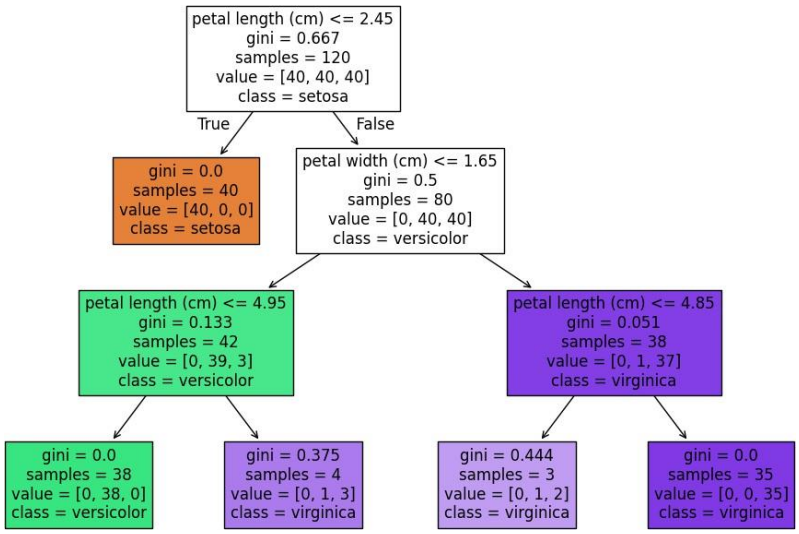
Output:

```
Confusion Matrix:
[[13  0  0]
 [ 0  9  2]
 [ 0  0  6]]
Precision: 0.9166666666666666
Recall: 0.9393939393939394
F1 Score: 0.919047619047619
```

Accuracy: 0.9666666666666667

Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	0.90	0.95	10
virginica	0.91	1.00	0.95	10
accuracy			0.97	30
macro avg	0.97	0.97	0.97	30
weighted avg	0.97	0.97	0.97	30



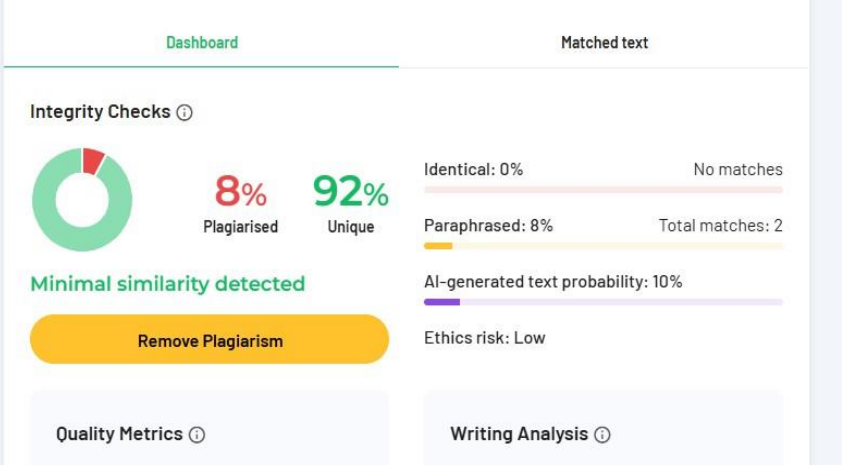
Output Analysis

Overfitting vs Underfitting Analysis

- **Overfitting: Occurs when the decision tree becomes too complex and learns the training data very well but performs poorly on new data.**
- **Underfitting: Occurs when the model is too simple and cannot capture the patterns in the data.**

This can be controlled by adjusting parameters such as maximum tree depth, minimum samples per leaf, or pruning techniques.

The decision tree classifier successfully predicted the class labels of the test dataset. The confusion matrix shows the number of correct and incorrect predictions. Precision, recall, and F1-score values indicate that the model performs well in classifying the data. Proper

	parameter tuning helps prevent overfitting and improves the model's generalization ability.
Link of student Github profile where lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git
Conclusion	The Decision Tree classifier was successfully implemented and evaluated using the Iris dataset. The performance was measured using the confusion matrix, precision, recall, and F1-score. The experiment also demonstrated how overfitting and underfitting can affect model performance and how they can be controlled through model tuning.
Plag Report (Similarity index < 12%)	
Date	02/03/26

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